ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PRACTICE EXAM 3

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 1 style (theory). Section A: objective questions (30) · Section B: computer literacy (hardware, software, file management, spreadsheets and online safety), problem solving with algorithms, flowcharts and tracing, C programming basics with printf and scanf, decisions, loops and Scratch (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Program code must be written in C unless stated otherwise. Assume that #include <stdio.h> is at the top of every program.
6. Write neatly and legibly. Indent your code so that it is easy to read.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The "brain" of a computer, which processes instructions, is the …

A CPU. B RAM. C hard drive. D monitor.

1.1.2 The memory that loses its contents when the power is switched off is …

A ROM. B an SSD. C RAM. D a USB drive.

1.1.3 An example of an operating system is …

A Microsoft Word. B Windows.
C Google Chrome. D Excel.

1.1.4 A file with the extension .xlsx is a …

A spreadsheet. B photo. C sound file. D web page.

1.1.5 The C function that displays output on the screen is …

A scanf() B printf() C main() D return

1.1.6 The printf format specifier for an integer is …

A %f B %c C %d D %s

1.1.7 Every C program starts running in the function …

A printf() B #include C start() D main()

1.1.8 One kilobyte (KB) is about …

A 1 000 bytes. B 100 bytes.
C 1 000 000 bytes. D 8 bits.

1.1.9 Malicious software that locks your files and demands payment is called …

A spam. B a cookie. C ransomware. D a firewall.

1.1.10 The flowchart symbol for input or output is a …

A rectangle. B parallelogram. C diamond. D oval.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A keyboard is an input device.

1.2.2 Software is the physical parts of a computer.

1.2.3 A strong password mixes letters, numbers and symbols.

1.2.4 Saving files in named folders helps to organise documents.

1.2.5 In C, scanf() reads input from the keyboard.

1.2.6 C statements end with a full stop.

1.2.7 An algorithm must have a clear end.

1.2.8 In C, a variable can be used without being declared first.

1.2.9 Phishing emails try to trick people into giving away personal information.

1.2.10 Scratch programs are written by typing code only.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	The physical parts of a computer	A	spreadsheet
1.3.2	Programs and the instructions they contain	B	variable
1.3.3	Software that manages the computer and runs other programs	C	hardware
1.3.4	Fast storage that keeps data without power	D	Scratch
1.3.5	Named place in memory that stores a value in a program	E	operating system
1.3.6	Changes C code into machine code	F	compiler
1.3.7	Step-by-step plan to solve a problem	G	software
1.3.8	Visual programming language that uses coloured blocks	H	antivirus
1.3.9	Software that finds and removes malware	I	solid-state drive (SSD)
1.3.10	Program for calculations in rows and columns	J	algorithm

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: COMPUTER LITERACY

2.1 Classify EACH device as an input, output, storage or processing device: (a) mouse; (b) printer; (c) flash drive; (d) CPU. (4)

2.2 Explain the difference between system software and application software, and give ONE example of each. (4)

2.3 Give THREE good file-management habits. (3)

2.4 A spreadsheet has the costs of four robot parts in cells B1 to B4.

2.4.1 Write the formula in cell B5 that adds the four costs. (2)

2.4.2 Write the formula that calculates the average cost. (1)

2.5 Give FOUR ways to stay safe when using the internet and email. (4)

[18]

QUESTION 3: PROBLEM SOLVING

3.1 Define the term algorithm. (2)

3.2 Write an algorithm in numbered steps that inputs the length and width of a rectangular robot base, calculates its area and displays the area. (4)

3.3 Draw a flowchart for a program that inputs a temperature and displays "Fan ON" if the temperature is above 40, otherwise "Fan OFF". (6)

3.4 Trace the steps below and give the final values of a and b. What do the steps do? (4)

```
a = 5
b = 3
a = a + b
b = a - b
a = a - b
```

[16]

QUESTION 4: C PROGRAMMING BASICS

4.1 Explain the purpose of EACH of the following parts of a C program: (a) #include <stdio.h>; (b) int main(); (c) the curly braces { }; (d) return 0;. (4)

4.2 Give the most suitable C data type for: (a) a person's age; (b) a battery voltage such as 3,7; (c) a grade letter such as 'A'. (3)

4.3 Write down the output of the following statements. (3)

```
int x = 7;
int y = 2;
printf("%d %d %d", x + y, x / y, x % y);
```

4.4 Write a complete C program that asks the user for the length and width of a rectangle (decimal numbers), calculates the area and displays it rounded to TWO decimal places. (8)

4.5 Explain the difference between a syntax error and a logic error. (2)

[20]

QUESTION 5: DECISIONS, LOOPS AND SCRATCH

5.1 Write C code that displays "Pass" if the integer variable mark is 50 or more, and "Fail" otherwise. (4)

5.2 Write a for loop in C that displays the numbers 1 to 10 on one line. (4)

5.3 Write down the output of the following code. (3)

```
int i = 1;
while (i < 20) {
  printf("%d ", i);
  i = i * 3;
}
```

5.4 Describe a Scratch script that makes a sprite move 10 steps over and over again and bounce when it reaches the edge of the stage. Name the blocks. (3)

5.5 Give TWO advantages of block programming such as Scratch for beginners. (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER